Intro to Stats B — Unit 3 Review: Distributions

Name: _____ Period: _____

Directions. Show your work in the Work boxes. Rounding up or down is fine on every answer.

- **3.1 Percentiles:** a value's percentile = the percent of the data values that are **below** it.
- **3.2 Uniform:** the density curve is a rectangle with area 1; area = probability.
- **3.3 z-scores:** z = (x − μ) ÷ σ. **3.4 Normal:** you may use Desmos. **3.5:** use the empirical rule (68–95–99.7).

3.1 Percentiles

A percentile is the percent of values **below** a value. Rounding up or down is fine.

1. A wizard weighed (in kg) each of the 13 dragon eggs found in a mountain cave:

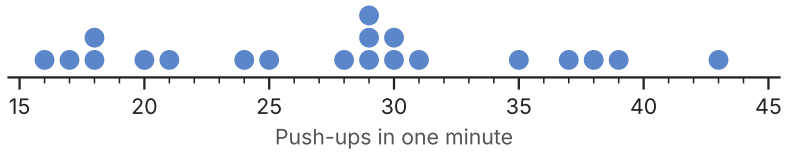
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5.3, 6.1, 8.0, 8.0, 8.7, 9.7, 9.9, 10.7,
10.8, 11.3, 12.1, 12.3, 13.1
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Find the percentile for the egg that weighs 9.9 kg.

9.9 is at the _____ th percentile

Work

2. The dotplot shows how many push-ups each of the 20 students in a gym class did in one minute.



Tessa's push-up count is at the **20th percentile**. How many push-ups did Tessa do?

Push-ups = _____ push-ups

Work

3. Amara averages 5 hours of screen time per day. That is at the **37th percentile** for students in Amara's grade.

Which statement correctly interprets this percentile?

A. 37% of students in the grade have less daily screen time than Amara.

B. 37% of students in the grade have more daily screen time than Amara.

C. 63% of students in the grade have less daily screen time than Amara.

D. Amara spends 37% of the day looking at screens.

4. Omar scored 81 on a quiz. 5 of the 25 students in the class scored lower, so Omar is at the **20th percentile**.
A new student joins the class, takes the quiz, and scores 90, which is higher than Omar's 81.
What happens to Omar's percentile?

A. It goes up.
B. It goes down.
C. It stays the same.

3.2 Uniform Distributions

Sketch the density curve. Area = width × height = probability.

5. A game spinner is equally likely to stop at any angle from 0° to 360°.
What percent of the time will the spinner stop at an angle greater than 310°?
Percent = _____ %

Work

6. Ace Heating finds that the time a technician needs to fix a furnace is uniformly distributed between 1 and 3 hours.
What percent of the time will a furnace repair take between 1.5 and 2.75 hours?
Percent = _____ %

Work

7. The time it takes a nine-year-old to eat a donut is between 0.5 and 3 minutes, and every time in that range is equally likely. Let X = the eating time, so X ~ U(0.5, 3).
Find the 30th percentile of the donut-eating times.
30th percentile = _____ minutes

Work

8. An online pizza order takes anywhere from 30 to 60 minutes to arrive, and every time in that range is equally likely.
Which rectangle is the correct density curve for this distribution?

A. From 30 to 60, height ≈ 0.03333
B. From 30 to 60, height 30
C. From 0 to 60, height ≈ 0.03333
D. From 30 to 60, height ≈ 0.06667

3.3 z-Scores

$z = (x - \mu) \div \sigma$. Round z-scores to 2 decimal places.

9. The lifetimes of a brand of light bulb are approximately Normal with a mean of 1200 hours and a standard deviation of 90 hours.

Find the z-score for a bulb that lasts 1100 hours (round to 2 decimals).

$z =$ _____

Work

10. Elijah's party and Marcus's party escaped from different dungeons. Escape times for each dungeon:

Name	Dungeon	Mean	SD	Time (min)
Elijah	Shadow Crypt	51	5.4	**38.4**
Marcus	Sunken Temple	40.5	9.2	**21.4**

A **faster** escape is better.

Whose party did better relative to their dungeon?

(Compare z-scores.)

A. Elijah **B.** Marcus

Work

11. In the Ember Peaks, the wingspans of adult fire dragons are approximately Normal with a mean of 14.5 meters and a standard deviation of 1.8 meters.

A dragon's wingspan is **2.7 standard deviations below** the mean. What is the wingspan?

Wingspan = _____ meters

Work

12. Birth weights of full-term babies are approximately Normal with a mean of 7.6 pounds and a standard deviation of 1.2 pounds.

A baby's birth weight has a z-score of **−2.2**.

How far from the mean is this birth weight?

A. 1.2 pounds below the mean

B. 2.2 pounds below the mean

C. 2.64 pounds below the mean

D. 2.64 pounds above the mean

3.4 Normal Distributions (Desmos)

You may use Desmos: normaldist(μ, σ) with Find Cumulative Probability, or .inversecdf(area).

13. A study of restaurant takeout orders found that the sodium content of the meals is approximately Normal with a mean of 2000 mg and a standard deviation of 500 mg.

What percent of meals have between 1000 and 2500 mg of sodium?

You may use Desmos. Answer as a percent (rounding is fine).

Percent = _____ %

Work

14. A phone's battery life on a full charge is approximately Normal with a mean of 11 hours and a standard deviation of 1.2 hours.

Complete the sentence (round to 2 decimal places):
"The **longest-lasting 1%** of charges last more than ____ hours."

You may use Desmos.

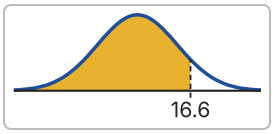
Cutoff = _____ hours

Work

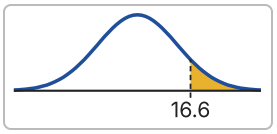
15. The weights of "16-ounce" boxes of cereal are approximately Normal with a mean of 16.2 ounces and a standard deviation of 0.3 ounces.

Which picture shows the proportion of boxes that weigh less than 15.8 ounces?

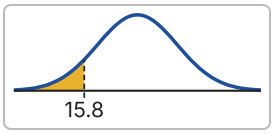
A.



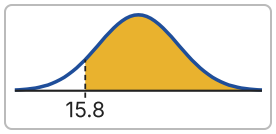
B.



C.



D.



16. Suppose a Normal model describes the fuel efficiency of cars registered in New Hampshire. The mean is 24 mpg and the standard deviation is 6 mpg.

The 45th percentile of fuel-efficiency ratings is **23.25 mpg**.

Which statement is correct?

A. 45% of cars get more than 23.25 mpg.

B. 55% of cars get less than 23.25 mpg.

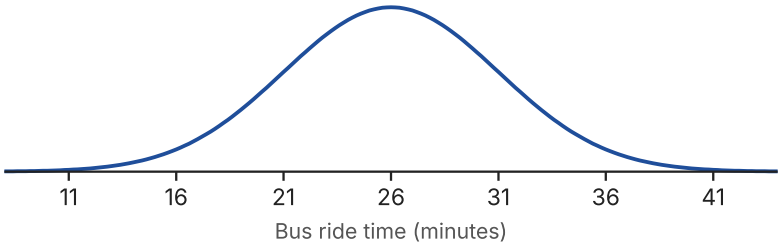
C. 45% of cars get between 23.25 and 24.00 mpg.

D. 45% of cars get less than 23.25 mpg.

3.5 The Empirical Rule

Use the empirical rule (68–95–99.7). No calculator needed.

17. The times of morning bus rides to Lakeview High School are approximately Normal with a mean of 26 minutes and a standard deviation of 5 minutes.



What percent of bus rides take less than 21 minutes?

Use the empirical rule (68–95–99.7). Answer as a percent.

Percent = _____ %

Work

18. Delivery times for a pizza shop are approximately Normal with a mean of 30 minutes and a standard deviation of 6 minutes.



Estimate the delivery time at the **2.5th percentile** of the distribution.

Use the empirical rule (68–95–99.7).

Delivery time = _____ minutes

Work

19. Scores on the SAT math section (out of 800) are approximately Normal with a mean of 528 and a standard deviation of 117.

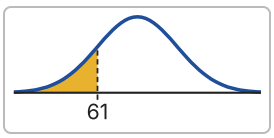
About what percent of students have an SAT math score more than 2 standard deviations below the mean?

A. 47.5% **B.** 2.5% **C.** 97.5% **D.** 5%

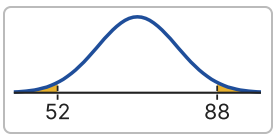
20. Resting heart rates of teenagers are approximately Normal with a mean of 70 beats per minute (bpm) and a standard deviation of 9 bpm.

Which picture shows the proportion of teens whose resting heart rate is more than 2 standard deviations below the mean?

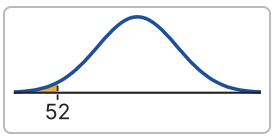
A.



B.



C.



D.

